

Complete SQL Mastery Syllabus

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Program Overview

This syllabus is designed to build strong expertise in SQL for database management, analytics, backend development, and enterprise applications.

Phases Included:

1. Database Fundamentals
2. SQL Basics
3. Advanced SQL Queries
4. Joins & Relationships
5. Functions & Procedures
6. Indexing & Optimization
7. Transactions & Security
8. SQL for Analytics
9. Real-Time Projects
10. Interview Preparation

Key Topics:

- Introduction to Databases
- RDBMS Concepts
- MySQL & PostgreSQL
- SQL Syntax
- CRUD Operations
- WHERE, ORDER BY, GROUP BY
- Aggregate Functions
- Joins
- Subqueries

- Views
- Stored Procedures
- Triggers
- Transactions
- ACID Properties
- Indexing
- Query Optimization
- Window Functions
- Common Table Expressions (CTE)

Hands-on Projects:

- Employee Database System
- Banking Database
- Inventory Management Database
- E-commerce Database
- Student ERP Database
- Sales Analytics Dashboard

Recommended Tools:

- MySQL Workbench
- PostgreSQL
- SQL Server
- Oracle Database
- DBeaver

Career Outcomes:

- SQL Developer
- Database Administrator
- Data Analyst
- Backend Developer
- Business Intelligence Analyst

Final Advice:

Practice SQL queries daily, master joins and optimization techniques, build database projects, and solve real-world analytical problems consistently.